

Generation Interconnection Request System Impact Study Report

For Queue Project Q16-0570

Tahkenitch 100 kV
April 22, 2017

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q16-0570. The Interconnection Customer proposes a 217.2 MW battery energy storage facility interconnecting at the Tahkenitch 100 kV substation in the Illinois area. The request was received on January 09, 2016. The study evaluated system intact and single contingency conditions using the regional planning case, including 277 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$64,307,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Gas / Micco 100 kV	Top Crop 230-100 kV XFMR	271.4	115	100.0
Westwood / Micco 100 kV L	Top Crop 230/100 kV Transformer	258.8	131	100.0
Astro / Pukele 100 kV	Micco / Westwood 100 kV	243.3	137	100.0
Pukele Substation / Top Crop 100 kV	Micco Substation / Westwood 100 kV Line	241.9	140	100.0
Astro / Sugar Grove 100 kV	Micco / Westwood 100 kV	238.8	135	100.0
Sugar Grove / Tahkenitch 100 kV	Micco / Westwood 100 kV	235.4	130	100.0
Gas / Amax Mines 100 kV Line	Top Crop 230 kV/100 kV Transformer Bank	226.2	120	100.0
Tahkenitch / Amax Mines 100 kV L	Top Crop 230 kV/100 kV Autotransformer	208.0	126	100.0
Top Crop 230/100kV XFMR	Micco / Westwood 100 kV	174.1	194	100.0
Chevron / Bent Mtn 100 kV	Chevron / Bent Mountain 100 kV Circuit 2	153.5	164	41.5
Westwood / Bent Mtn 100 kV Circuit 2	Bent Mtn / Westwood 100 kV Line 1	151.7	146	39.8
Bent Mtn / Westwood 100 kV Line 1	Bent Mountain / Westwood 100 kV Line 2	142.4	156	39.8
Bent Mtn / Chevron 100 kV Cir. 2	Chevron Substation / Bent Mtn 100 kV Line	141.4	178	41.5
Ligonier / Cypress Crk 100 kV	Dekalb / Jones Chapel 500 kV	137.5	232	5.4
Chevron / Fort St Vrain 100 kV Line 1	Fort St Vrain / Chevron 100 kV	137.2	206	41.7
Chevron / Ft St Vrain 100 kV Ckt 2	Chevron / Ft St Vrain 100 kV	133.5	211	41.7

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Big Oaks / Cypress Creek 100 kV L	Dekalb / Jones Chapel 500 kV L	128.1	241	5.4
Cuba City / Farmersville 100 kV	Dekalb / Jones Chapel 500 kV	114.8	129	5.4
Paliades 230/1 kV Autotransformer	Flood St / Cook 500 kV	114.7	806	9.7
Palisades 500 kV/1 kV XFMR	Cook / Flood St 500 kV	114.2	810	9.7
Dairymans Substation / Pueblo Xtra 230 kV Circuit 2	Dairymans / Pueblo Xtra 230 kV Line 1	113.6	846	6.7
Pueblo Xtra / Dairymans 230 kV	Dairymans / Pueblo Xtra 230 kV Circuit 2	108.6	885	6.7
Big Oaks / Laureles 100 kV	Dekalb / Jones Chapel 500 kV	107.5	265	5.4
Twin County Ranch / Richland 230 kV Line	Trego Substation / Swan Vly 500 kV	105.0	649	8.1
Laotto / Golden Gate 100 kV L	Dekalb / Jones Chapel 500 kV	104.8	193	5.4

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Micco / Gas 100 kV	Rebuild line with higher-capacity conductor	\$2,230,000	100% (\$2,230,000)
Westwood / Micco 100 kV L	Rebuild line with higher-capacity conductor	\$1,075,000	100% (\$1,075,000)
Pukele / Astro 100 kV	Rebuild line with higher-capacity conductor	\$7,137,000	100% (\$7,137,000)
Pukele / Top Crop 100 kV	Rebuild line with higher-capacity conductor	\$3,927,000	100% (\$3,927,000)
Astro / Sugar Grove 100 kV	Rebuild line with higher-capacity conductor	\$4,331,000	100% (\$4,331,000)
Tahkenitch / Sugar Grove 100 kV	Rebuild line with higher-capacity conductor	\$2,264,000	100% (\$2,264,000)
Gas / Amax Mines 100 kV Line	Rebuild line with higher-capacity conductor	\$10,247,000	100% (\$10,247,000)
Tahkenitch / Amax Mines 100 kV	Rebuild line with higher-capacity conductor	\$7,623,000	100% (\$7,623,000)
Top Crop 230/100kV XFMR	Replace transformer with higher-rated unit	\$5,984,000	100% (\$5,984,000)
Bent Mountain / Chevron 100 kV Circuit 1	Rebuild line with higher-capacity conductor	\$1,797,000	100% (\$1,797,000)
Bent Mtn / Westwood 100 kV Cir. 2	Rebuild line with higher-capacity conductor	\$4,066,000	100% (\$4,066,000)
Westwood / Bent Mountain 100 kV line	Rebuild line with higher-capacity conductor	\$4,980,000	100% (\$4,980,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Bent Mtn / Chevron 100 kV Cir. 2	Rebuild line with higher-capacity conductor	\$1,521,000	100% (\$1,521,000)
Ligonier / Cypress Crk 100 kV	Rebuild line with higher-capacity conductor	\$7,490,000	3% (\$222,000)
Chevron / Fort St Vrain 100 kV Line 1	Rebuild line with higher-capacity conductor	\$1,792,000	100% (\$1,792,000)
Chevron / Ft St Vrain 100 kV Ckt 2	Rebuild line with higher-capacity conductor	\$1,625,000	100% (\$1,625,000)
Big Oaks / Cypress Creek 100 kV L	Rebuild line with higher-capacity conductor	\$8,302,000	3% (\$246,000)
Farmersville / Cuba City 100 kV	Rebuild line with higher-capacity conductor	\$6,522,000	3% (\$194,000)
Palsades 230-1 kV Transformer	Replace transformer with higher-rated unit	\$10,970,000	3% (\$323,000)
Palisades 500 kV/1 kV XFMR	Replace transformer with higher-rated unit	\$21,307,000	3% (\$628,000)
Dairymans / Pueblo Xtra 230 kV #2	Rebuild line with higher-capacity conductor	\$40,161,000	2% (\$903,000)
Dairymans / Pueblo Xtra 230 kV	Reconductor line	\$26,362,000	2% (\$593,000)
Big Oaks / Laureles 100 kV	Reconductor line	\$1,169,000	3% (\$35,000)
Twin County Ranch / Richlad Substation 230 kV	Reconductor line	\$17,020,000	3% (\$508,000)
Golden Gate / Laotto 100 kV	Reconductor line	\$1,894,000	3% (\$56,000)
	Total allocated to project		\$64,307,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.